

Naga Pranav Puppala

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EDUCATION

The University of Texas at Dallas <i>Master of Science, Computer Science - Intelligent Systems</i>	Richardson, TX Aug. 2026 – Dec. 2027
The University of Texas at Dallas <i>Bachelor of Science, Computer Science, GPA: 3.72</i>	Richardson, TX Jan. 2023 – May 2026

EXPERIENCE

Software Engineer Intern (Capstone Project) <i>UT Southwestern Medical Center</i>	Feb. 2026 – May 2026 Dallas, TX
- Built a hybrid RAG pipeline using LangChain, ChromaDB, BM25, and cross-encoder reranking to generate clinically accurate, persona-aware Major Depressive Disorder patient responses for psychiatric interview training - Designed a synthetic persona generation pipeline that extracted structured clinical profiles from therapy transcripts using Groq and LLaMA, scored multi-dimensional patient similarity via ChromaDB, and synthesized blended MDD personas for real-time conversational simulation - Developed a real-time psychiatric patient simulation system with structured behavioral metadata tagging (mood, affect, and symptom) producing both natural dialogue and clinical JSON output	

PROJECTS

ExpenseTracker <i>Django REST, React, JWT Authentication</i>	Aug. 2025 – Oct. 2025
- Built full-stack expense management application with Django REST API backend featuring JWT authentication, user registration, login, and CRUD operations for expense tracking - Implemented filtering and pagination systems with real-time search, category-based filtering, and dashboard analytics with trend visualization using Recharts - Designed UI with React and Tailwind CSS including dark theme design, form validation, and user profile management with password change functionality	
KnowledgeAtlas <i>Django REST, React, LangChain, Celery, Docker</i>	Jun. 2026 – Jul. 2026
- Built a full-stack RAG (Retrieval-Augmented Generation) application with a Django REST backend and React frontend, using LangChain and the Groq API to generate grounded, streamed, citation-backed answers from user-uploaded PDFs - Architected an asynchronous document ingestion pipeline with Celery and Redis, decoupling PDF parsing/chunking/embedding from the request cycle and containerizing the full stack (backend, worker, Redis, frontend) with Docker Compose for one-command local deployment - Designed a vector retrieval system using ChromaDB and sentence-transformers, calibrating similarity thresholds against real test data and building an evaluation harness that verified retrieval accuracy at a 100% pass rate across a curated question set	
Resume Analyzer <i>React, TypeScript, Puter.js</i>	Jul. 2025 – Aug. 2025
- Created web-based resume analysis platform integrating Puter.js for user authentication, file uploads, AI feedback, and persistent key-value storage - Designed end-to-end flow where users upload resumes, receive structured ATS evaluations from Claude, and view personalized feedback in dynamic UI - Created custom React components including interactive score gauges, tip accordions, and resume preview cards to enhance user experience	

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL, PostgreSQL, JavaScript, HTML, CSS
Frameworks: React, Next.js, Django, Django REST Framework, Scikit-Learn, TensorFlow
Developer Tools: Git, VS Code, GitHub, Docker, Claude Code, ChatGPT, Perplexity AI,
Libraries: Pandas, NumPy, Matplotlib, Axios, Prisma, Tailwind CSS, Recharts, Zustand, LangChain, ChromaDB, Celery, Redis